GENERATIVE MODELING · FLOW MATCHING

Flow Matching & Optimal Transport

From normalizing flows to conditional flow matching

$$\partial_t p_t + \nabla \cdot \left(p_t \, v_t\right) = 0$$

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Overview

01 **Normalizing Flows & CNFs**
Discrete flows, the Jacobian bottleneck, and continuous-time flows.

02 **Flow Matching**
The continuity equation and vector-field regression.

03 **Conditional Flow Matching**
Conditional paths, marginalization, and the CFM theorem.

04 **Gaussian Paths & the Closed-Form Field**
The canonical affine flow and its exact velocity.

05 **Diffusion as a Special Case**
Recovering VE / VP and the probability-flow ODE.

06 **Optimal Transport Paths**
Straight-line transport and more efficient sampling.

07 **Results from the Paper**
ImageNet & CIFAR-10, sampling efficiency, super-resolution.

PART 01

Normalizing Flows & CNFs

The core problem · Discrete flows · Continuous flows · The training cost

The Core Problem in Generative Modeling

Goal: Learn a model of a complex, high-dimensional data distribution $p_1(x)$ over data x.

The fundamental challenge: real-world data (images, audio, molecules) lives on complex, high-dimensional manifolds that are intractable to model directly.

1. **Start from a simple base**
   A tractable distribution p_0, e.g. a standard Gaussian $\mathcal{N}(0, I)$.
2. **Learn a transformation**
   A map that pushes the base p_0 forward onto the data p_1.
3. **Use it both ways**
   The same transformation gives sampling and density estimation.

Central question: how do we parameterize and learn this transformation efficiently?

Discrete Normalizing Flows

APPROACH

Learn an invertible, differentiable map ϕ so that x = ϕ(z) for base noise z ∼ p_0.

$$\phi : \mathbb{R}^d \to \mathbb{R}^d, \quad x = \phi(z), \quad z \sim p_0$$

CHANGE OF VARIABLES

$$p_1(x) = p_0\big(\phi^{-1}(x)\big) \left|\det \frac{\partial \phi^{-1}}{\partial x}\right|$$

1. **Invert**
   Compute z = $\phi^{-1}(x)$ from the data point.
2. **Evaluate base density**
   Compute $p_0(z)$ under the simple prior.
3. **Add log-Jacobian**
   Correct for the volume change of the map.

TRAINING OBJECTIVE · MAXIMUM LIKELIHOOD

$$\max_{\theta} \quad \mathbb{E}_{x \sim p_1}\big[\log p_1(x)\big]$$

[4] Rezende & Mohamed, 2015 · [5] Dinh et al., 2017 (RealNVP)

The Bottleneck of Discrete Flows

The Jacobian determinant is the central computational bottleneck.

For a general dense Jacobian, exact computation costs $O(d^3)$ — infeasible in high dimensions. Every practical family buys tractability by restricting the architecture.

Method	Trick	Limitation
RealNVP / Glow	Coupling layers	Restricted expressivity
Autoregressive (MAF, IAF)	Triangular Jacobian	Slow sampling or slow training
Residual flows	Trace estimator	Approximate likelihood

Fundamental tension: tractability of the Jacobian requires restricting the architecture, which inherently limits the expressivity of the learned map.

[5] RealNVP · [6] Glow · [7] MAF / IAF

Continuous Normalizing Flows (CNFs)

FLOW ODE

$$\frac{d}{dt}\phi_t(x) = v_t\big(\phi_t(x)\big), \qquad \phi_0(x) = x$$

- ϕ_t is the flow map at time $t \in [0, 1]$.
- $v_t(\cdot\,;\theta)$ is a learned, time-dependent vector field, parameterized by a neural network.

Key idea: replace the discrete invertible map with a continuous-time dynamical system governed by an ODE.

DENSITY TRANSPORT

$$p_t = [\phi_t]_* \, p_0$$

SAMPLING

Integrate the ODE forward from $t = 0$ to $t = 1$, starting from $z \sim p_0$.

[2] Chen et al., 2018 — Neural Ordinary Differential Equations

Density Evaluation in CNFs

Change of variables still applies at every time t:

$$p_t(x) = p_0\big(\phi_t^{-1}(x)\big)\,\left|\det \frac{\partial \phi_t^{-1}}{\partial x}\right|$$

Two compounding difficulties

1. NO CLOSED FORM FOR Φ_t

The flow map must be obtained by numerically integrating the ODE, which is costly at test time.

2. JACOBIAN IS STILL $O(D^3)$

Exact density evaluation still needs the full Jacobian, which scales poorly with dimension.

These issues motivate a smarter way to track how density changes along the flow.

The Instantaneous Change of Variables

Key insight: instead of tracking the global volume change via the determinant, track the instantaneous rate of change via the divergence.

INSTANTANEOUS CHANGE OF VARIABLES (LIOUVILLE)

$$\log p_1(x_1) = \log p_0(x_0) - \int_0^1 \operatorname{div}\big(v_t(x_t)\big)\, dt$$

where $x_t = \phi_t(x_0)$ is the trajectory of x_0 under the flow.

GLOBAL VS INSTANTANEOUS

- Global: a determinant volume ratio at time t.
- Instantaneous: accumulated local expansion / compression.

The divergence is the trace of the Jacobian: $\operatorname{div}(v_t) = \operatorname{tr}(\nabla_x v_t)$ costs $O(d^2)$ exactly — but, crucially, it is amenable to cheap stochastic estimation.

Practical Density Computation in CNFs

AUGMENTED ODE SYSTEM

Jointly integrate the state and the log-density along the trajectory.
Boundary: $x_0 \sim p_0$, $f(1) = 0$.

$$\dot{x}_t = v_t(x_t)$$

$$\dot{f}(t) = -\,\mathrm{div}\big(v_t(x_t)\big)$$

OUTPUT (INTEGRATE T = 1 → 0)

$$\log p_1(x_1) = \log p_0(x_0) - f(0)$$

Stochastic trace estimator (Hutchinson): an unbiased estimate of the divergence.

$$\mathrm{tr}(\nabla_x v_t) \;\approx\; \epsilon^\top (\nabla_x v_t)\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

Reduces the per-sample cost of the divergence from $O(d^2)$ to $O(d)$.

[2] Chen et al., 2018 (Neural ODEs) · [3] Grathwohl et al., 2018 (FFJORD)

Why CNFs Remain Difficult to Train

Despite their theoretical elegance, Continuous Normalizing Flows face serious practical obstacles in training-time cost.

Issue	Consequence
ODE solve per sample	Many network function evaluations per gradient step
Adjoint method for gradients	Numerically unstable; memory-intensive
Divergence computation	Needs a Jacobian–vector product even with Hutchinson
Sequential time integration	Cannot be parallelized across time steps

Net effect: CNF training is orders of magnitude slower than discrete-flow training on the same architecture, making scaling to high-resolution data challenging.

CNFs — Summary

WHAT CNFS ACHIEVE

✓ Flexible density modeling with no architectural constraints on the vector field v_t.

✓ Exact — or stochastically approximate — likelihood evaluation.

✓ Continuous interpolation between the base p_0 and the data p_1.

THE REMAINING BOTTLENECK

Component	Cost
Learn	Vector field $v_t(\cdot\ ;\theta)$ via MLE + adjoint
Flow	Implicit; obtained by ODE integration
Likelihood	Trajectory + divergence integral

Conclusion: CNFs are expressive but expensive — training simulates the ODE, coupling forward and backward passes in a costly, unstable way.

PART 02

Flow Matching

Continuity equation · Vector-field regression · From CNF to FM

The Key Observation Behind Flow Matching

Critical observation: ODE simulation is not needed during training — if we can directly supervise the vector field.

PROPOSED SHIFT IN PERSPECTIVE

1. **Define a path**
   Choose a probability path p_t interpolating $p_0 \to p_1$ analytically.

2. **Derive the target field**
   Get the vector field u_t that generates the path via the continuity equation.

3. **Regress**
   Train $v_t(\cdot\,;\theta)$ to match u_t — no ODE solve required.

PARADIGM SHIFT

Simulation-based MLE
(CNF)

↓

Supervised vector-field regression
(Flow Matching)

This reframing converts an expensive, numerically unstable optimization into a stable, scalable supervised learning problem.

Flow Matching — Formulation

GOVERNING EQUATION — CONTINUITY

$$\partial_t p_t + \nabla \cdot (p_t\, u_t) = 0$$

Any pair (p_t, u_t) satisfying this equation defines a valid generative flow from p_0 to p_1.

FLOW MATCHING OBJECTIVE

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,\, x_t \sim p_t} \left\| v_t(x_t; \theta) - u_t(x_t) \right\|^2$$

ADVANTAGES OVER CNF TRAINING

- ✓ No ODE solve during training.
- ✓ No adjoint-method gradients.
- ✓ Numerically stable and highly parallelizable.
- ✓ Scales to large neural-network architectures.

Comparing the Three Paradigms

A unified view of flow-based generative models.

Method	What is learned	Training cost	Inference cost
Discrete NF	Invertible map ϕ	Jacobian determinant $O(d^3)$	Single forward pass
CNF	Vector field v_t	ODE solve + divergence integral	ODE integration
Flow Matching	Vector field v_t	Regression (MSE)	ODE integration

PROGRESSION OF IDEAS

- → Discrete NF — exact, but architecturally constrained.
- → CNF — flexible, but training is simulation-based and costly.
- → Flow Matching — flexible, and training reduces to regression.

In one line: Flow Matching inherits the expressiveness of CNFs while replacing simulation-based training with a simple, stable supervised objective.

Open Questions and Next Steps

WHAT WE HAVE ESTABLISHED

✓ Flow Matching frames generative modeling as vector-field regression.

✓ The continuity equation guarantees the learned flow transports $p_0 \to p_1$.

TWO OPEN QUESTIONS REMAIN

1. **How do we construct the path p_t?**
   We need an analytically tractable interpolation between p_0 and p_1.

2. **How do we derive the field u_t?**
   The marginal u_t is generally intractable — it requires integrating over all data.

NEXT: CONDITIONAL FLOW MATCHING

Construct p_t and u_t conditionally on individual data points, then marginalize analytically to recover a tractable training objective.

Conditional Flow Matching: Problem Setup

DATA

$$x_1 \sim q(x)$$

where q(x) is the unknown data distribution.

Constraint: we only have sample access — no density evaluation of q.

Goal: learn a generative model of the data distribution q(x).

Probability Paths and the Flow View

DEFINE A PATH

$$p_t(x), \qquad t \in [0, 1]$$

REQUIREMENTS

- $p_0 = p$, a simple base distribution, e.g. $\mathcal{N}(0, I)$.
- $p_1 \approx q$, the target data distribution.

> **Interpretation:** a continuous transform from noise to data.

THE FLOW VIEW

Learn a vector field $u_t(x)$, tied to the path by the continuity equation:

$$\partial_t p_t + \nabla \cdot \left(p_t\, u_t\right) = 0$$

- u_t transports probability mass through space.
- It defines the evolution of the probability path p_t.

The Flow Matching Objective

LEARN A NEURAL FIELD

$$v_t(x;\theta)$$

LOSS

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,x}\left|\left|v_t(x) - u_t(x)\right|\right|^2$$

INTERPRETATION

A supervised regression problem.

Target: $v_t \approx u_t$.

OPTIMUM BEHAVIOR (IF $V_t = U_t$)

- ✓ The learned flow exactly reproduces the probability path p_t.
- ✓ The final distribution matches the data: $p_1 \approx q$.

Why This Is Attractive

COMPARED TO CNFS

- ✓ No likelihood computation.
- ✓ No ODE solving required during training.
- ✓ No divergence estimation.

CORE BENEFIT

Reduces generative modeling to a simple regression objective.

The Core Challenge

WHAT WE NEED

$$p_t(x)$$

$$u_t(x)$$

THE ISSUES

✗ q(x) is unknown

We only have samples, not the density function.

✗ No closed-form p_t

Cannot easily compute the marginal probability path.

✗ No direct access to u_t

The target vector field is intractable to compute globally.

Key Insight: Go Conditional

USE CONDITIONAL OBJECTS

$$p_t(x \mid x_1) \qquad\qquad u_t(x \mid x_1)$$

Replace global construction → conditional construction

PART 03

Conditional Flow Matching

Conditional paths · Marginalization · The CFM theorem

Conditional Probability Paths

For each data sample x_1: define a conditional probability path $p_t(x \mid x_1)$ — one path per sample.

BOUNDARY CONDITIONS

T = 0 (NOISE)

$$p_0(x \mid x_1) = p(x)$$

the simple base distribution.

T = 1 (DATA)

$$p_1(x \mid x_1) \approx \mathcal{N}(x_1,\ \sigma_{\min}^2 I)$$

a tight Gaussian around the sample.

Interpretation: noise → a local Gaussian centered at x_1

Marginal Path Construction

MARGINAL PATH

$$p_t(x) = \int p_t(x \mid x_1)\, q(x_1)\, dx_1$$

AT T = 1

$$p_1(x) \approx q(x)$$

INTERPRETATION

The marginal path is a mixture of the per-sample conditional paths, weighted by the data distribution q.

Conditional Vector Fields

Each conditional path has an associated vector field that generates it:

$$u_t(x \mid x_1)$$

It satisfies the conditional continuity equation:

$$\partial_t\, p_t(x \mid x_1) + \nabla \cdot \big(p_t(x \mid x_1)\, u_t(x \mid x_1)\big) = 0$$

The Marginal Vector Field

DEFINITION — WEIGHTED AGGREGATION OF CONDITIONAL FIELDS

$$u_t(x) = \int u_t(x \mid x_1) \, \frac{p_t(x \mid x_1)\, q(x_1)}{p_t(x)} \, dx_1$$

EQUIVALENT FORM

$$u_t(x) = \mathbb{E}_{x_1 \sim p(x_1 \mid x, t)} \big[u_t(x \mid x_1) \big]$$

A posterior-weighted average over candidate data points x_1.

MEANING

- Many x_1 can explain the current state x.
- Each x_1 suggests its own direction $u_t(x \mid x_1)$.
- Combine them with posterior weights $p(x_1 \mid x, t)$.

Why This Aggregation Is Non-Trivial

NOT VALID

- ✗ Uniform averaging of the conditional fields.
- ✗ Plain (unweighted) simple averaging.

REQUIREMENT

The aggregated field must preserve the probability flow — the flux — of the underlying paths.

Key Result: The Marginal Field Works

THE MARGINAL FIELD SATISFIES THE CONTINUITY EQUATION

$$\partial_t p_t + \nabla \cdot \big(p_t\, u_t\big) = 0$$

CONCLUSION

The posterior-weighted aggregation of the tractable conditional fields generates exactly the marginal path p_t.

Proof — Setup

STEP 1 · START WITH THE MARGINAL PATH

$$p_t(x) = \int p_t(x \mid x_1)\, q(x_1)\, dx_1$$

STEP 2 · DEFINE THE AGGREGATED FLUX

$$p_t(x)\, u_t(x) = \int p_t(x \mid x_1)\, q(x_1)\, u_t(x \mid x_1)\, dx_1$$

Next: take the divergence of both sides — it moves inside the integral.

Proof — Taking the Divergence

DERIVATION

$$\begin{aligned}\nabla\cdot(p_t u_t) &= \int \nabla\cdot\big(p_t(x\mid x_1)\,u_t(x\mid x_1)\big)\,q(x_1)\,dx_1\\ &= \int \big(-\partial_t\,p_t(x\mid x_1)\big)\,q(x_1)\,dx_1\\ &= -\partial_t\int p_t(x\mid x_1)\,q(x_1)\,dx_1 = -\partial_t\,p_t(x)\end{aligned}$$

RESULT

Rearranging gives the continuity equation $\partial_t p_t + \nabla\cdot(p_t u_t) = 0$.

Why Conditional Flow Matching

INTRACTABLE — THE MARGINAL OBJECTIVE

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,x}\left\|v_t(x) - u_t(x)\right\|^2$$

↓ $u_t(x)$ is hard to compute

TRACTABLE — THE CONDITIONAL OBJECTIVE

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,\,x_1,\,x}\left\|v_t(x) - u_t(x \mid x_1)\right\|^2$$

$u_t(x \mid x_1)$ is known by design

The Optimal Flow Matching Theorem

Theorem. The two objectives have identical gradients with respect to the parameters θ:

$$\nabla_\theta \, \mathcal{L}_{\mathrm{FM}}(\theta) = \nabla_\theta \, \mathcal{L}_{\mathrm{CFM}}(\theta)$$

PROOF SKETCH

- Expand the squared norm in L_CFM.
- Use $E[\, u_t(x \mid x_1) \mid x \,] = u_t(x)$ — the posterior weighting appears automatically.
- The target-norm term is independent of θ.

CONSEQUENCE

Optimizing the tractable conditional loss exactly optimizes the intractable marginal loss.

PART 04

Gaussian Paths & the Closed-Form Field

Affine flow · Theorem 3 · The CFM objective

Conditional Probability Paths (Gaussian)

A TRACTABLE GAUSSIAN CONDITIONAL PATH

$$p_t(x \mid x_1) = \mathcal{N}\big(x;\ \mu_t(x_1),\ \sigma_t^2 I\big), \qquad \mu_t(x_1) = t\, x_1, \quad \sigma_t = 1 - (1 - \sigma_{\min})\, t$$

AT T = 0

$$p_0(x \mid x_1) = \mathcal{N}(x;\ 0,\ I)$$

begins as the simple base distribution.

AT T = 1

$$p_1(x \mid x_1) = \mathcal{N}\big(x;\ x_1,\ \sigma_{\min}^2 I\big)$$

a tight Gaussian around the data sample.

Each path transports pure noise toward a local Gaussian centered at the sample x_1.

Non-Uniqueness of Vector Fields

A probability path fixes how density changes — but it does ***not*** uniquely determine the velocity field.

$$\partial_t p_t(x) = -\nabla \cdot \big(p_t(x)\, u_t(x)\big) \qquad \Longrightarrow \qquad u'_t(x) = u_t(x) + w_t(x), \;\; \nabla \cdot \big(p_t w_t\big) = 0$$

SAME PATH

Continuity sees only the divergence of the flux.

EXTRA COMPONENTS

Divergence-free flux circulates mass without changing p_t.

NEED A CONVENTION

Pick a canonical field that is simple and computable.

Motivation: CFM pairs each Gaussian path with the simplest canonical vector field, not any equivalent field.

Canonical Flow Construction

DEFINE AN AFFINE FLOW

$$\psi_t(x) = \sigma_t(x_1)\, x + \mu_t(x_1)$$

PROPERTY

$$x \sim \mathcal{N}(0, I) \quad \Longrightarrow \quad \psi_t(x) \sim \mathcal{N}\big(\mu_t(x_1),\ \sigma_t^2(x_1)\, I\big)$$

The flow directly constructs the Gaussian path.

Vector Field from the Flow

The flow induces a vector field through its own time derivative:

$$\frac{d}{dt}\psi_t(x) = u_t\big(\psi_t(x) \mid x_1\big)$$

GOAL

Find a closed-form expression for the conditional field $u_t(x \mid x_1)$.

Key Result (Theorem 3)

$$u_t(x \mid x_1) = \frac{\dot{\sigma}_t(x_1)}{\sigma_t(x_1)}\big(x - \mu_t(x_1)\big) + \dot{\mu}_t(x_1)$$

SCALING TERM

The first term expands or contracts points relative to the mean, governed by the rate of change of σ_t.

TRANSLATION TERM

The second term shifts the whole distribution in space, following the drift of the mean μ_t.

Proof Sketch — Deriving the Field

Differentiate the affine flow, then re-express in terms of the current position.

$$\psi_t(x) = \sigma_t(x_1)\, x + \mu_t(x_1)$$

$$\frac{d}{dt}\psi_t(x) = \dot{\sigma}_t(x_1)\, x + \dot{\mu}_t(x_1)$$

$$y = \psi_t(x) \;\Rightarrow\; x = \frac{y - \mu_t(x_1)}{\sigma_t(x_1)}$$

$$u_t(x \mid x_1) = \frac{\dot{\sigma}_t(x_1)}{\sigma_t(x_1)}\big(x - \mu_t(x_1)\big) + \dot{\mu}_t(x_1)$$

Since the current position is arbitrary, the identity holds for every x.

The CFM Objective

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}_{t,\, x_1,\, x_0} \left\| v_t\big(\psi_t(x_0)\big) - \frac{d}{dt}\psi_t(x_0) \right\|^2$$

KEY IDEA

Train directly on the known velocities of the constructed paths.

The Big Picture

STEP 1

Define conditional Gaussian paths

For each sample x_1, construct a tractable path $p_t(x \mid x_1)$ from noise toward the sample.

STEP 2

Derive closed-form vector fields

The conditional velocity $u_t(x \mid x_1)$ is exactly computable along the constructed flow.

STEP 3

Train by regression

Fit $v_t(t, x)$ to the known velocities with the CFM loss — no likelihood integration.

RESULT

Conditional construction turns generative modeling into supervised vector-field learning — a scalable training procedure.

One-Line Takeaway

Gaussian conditional paths give a simple affine flow with an exactly computable velocity field.

$$\psi_t(x_0) = \sigma_t\, x_0 + t\, x_1, \qquad u_t(x \mid x_1) = \frac{x_1 - (1 - \sigma_{\min})\, x}{1 - (1 - \sigma_{\min})\, t}$$

SIMPLE PATH

The conditional transport is affine in the base sample.

EXACT TARGET

The velocity target is available without simulation.

EFFICIENT TRAINING

Flow Matching becomes direct vector-field regression.

PART 05

Diffusion as a Special Case

General formulation · VE & VP · Probability-flow ODE

Flow Matching with Diffusion Paths

Flow Matching is a framework for the **design of probability paths** — and diffusion is one instance.

FLEXIBLE BY DESIGN

Flow Matching allows flexible design of the probability path connecting noise to data.

RECOVERS DIFFUSION

Standard diffusion processes arise as a special case of this design space.

GOES BEYOND DIFFUSION

It also enables non-diffusion paths, such as Optimal Transport.

General Formulation

CONDITIONAL GAUSSIAN PATHS

$$p_t(x \mid x_1) = \mathcal{N}\big(x \mid \mu_t(x_1),\ \sigma_t^2(x_1)\, I\big)$$

DESIGN CHOICES

- Mean schedule $\mu_t(x_1)$.
- Standard-deviation schedule $\sigma_t(x_1)$.

CORE REQUIREMENTS

- Both must be smooth functions of t.
- Boundary: $t = 0$ is noise, $t = 1$ is data.

$\Rightarrow$ a fully flexible design space

Key Idea: Direct Probability Paths

Flow Matching works directly with probability paths.

No need to derive them from diffusion SDEs.

RECOVER

Reproduce standard diffusion paths exactly.

DESIGN

Build entirely new paths, e.g. Optimal Transport.

Diffusion Paths as a Special Case

Diffusion models define a reverse process:

Noise → **Data**

THESE CORRESPOND TO SPECIFIC CHOICES OF

$$\text{mean} \quad \mu_t(x_1) \qquad \qquad \text{std. dev.} \quad \sigma_t(x_1)$$

Result: the conditional paths are exactly Gaussian.

Example: Variance Exploding (VE)

PATH

$$p_t(x \mid x_1) = \mathcal{N}\big(x \mid x_1,\ \sigma_{1-t}^2\, I\big)$$

PARAMETERS

$$\mu_t(x_1) = x_1, \quad \sigma_t(x_1) = \sigma_{1-t}$$

VECTOR FIELD

$$u_t(x \mid x_1) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}\,(x - x_1)$$

Points move toward x_1 with a time-dependent scaling.

Example: Variance Preserving (VP)

PATH

$$p_t(x \mid x_1) = \mathcal{N}\big(x \mid \alpha_{1-t}\, x_1,\ (1 - \alpha_{1-t}^2)\, I\big)$$

PARAMETERS

$$\mu_t(x_1) = \alpha_{1-t}\, x_1, \quad \sigma_t(x_1) = \sqrt{1 - \alpha_{1-t}^2}$$

VECTOR FIELD

A more complex form, depending on x, x_1, and time t.

Result: matches standard VP diffusion dynamics.

Same Dynamics as the Probability-Flow ODE

These conditional vector fields

$$u_t(x \mid x_1)$$

exactly match the **probability-flow ODE** of diffusion models.

Same dynamics as: score-based diffusion (Song et al., 2021) — the deterministic sampler is one and the same ODE.

[9] Song et al., 2021 — Score-Based Generative Modeling through SDEs

Same Field, Different Training Objective

Flow Matching uses the same vector field as diffusion — but optimizes a different objective.

DIFFUSION MODELS

Score Matching

FLOW MATCHING

Vector-Field Regression

FLOW MATCHING PROVIDES

- ✓ More stable training.
- ✓ More robust optimization.

Limitation of Diffusion, Advantage of FM

Theoretical issue: diffusion processes do not reach exact noise in finite time — in practice they must rely on approximations.

THE FLOW MATCHING FRAMEWORK ENABLES

1. **Exact path design**
   Specify the probability path directly, with no SDE approximation.
2. **More flexibility**
   Freely choose mean and variance schedules.
3. **Better alternatives**
   Adopt geometry-driven paths such as Optimal Transport.

PART 06

Optimal Transport Paths

Straight-line transport · OT flow map · Diffusion vs OT

Two Views of the Vector Field

EULERIAN VIEW

$$u_t(x \mid x_1)$$

defined globally at any point x.

LAGRANGIAN VIEW

$$\frac{d}{dt}\psi_t(x_0)$$

velocity along a specific trajectory.

The Key Identity

$$u_t\big(\psi_t(x_0) \mid x_1\big) = \frac{d}{dt}\psi_t(x_0)$$

Why Training Uses the Derivative

INSTEAD OF THE GLOBAL FIELD

$$u_t(x)$$

USE THE TRAJECTORY VELOCITY

$$\frac{d}{dt}\psi_t(x_0)$$

WHY — THREE CHEAP STEPS

1. **Sample noise**
   Draw $x_0 \sim \mathcal{N}(0, I)$.
2. **Compute position**
   Evaluate $\psi_t(x_0)$ along the path.
3. **Differentiate**
   Read the velocity off directly — no global solve.

From Flow to Field — the Algebra

Differentiate the OT flow, invert for the base point, and substitute.

$$\psi_t(x_0) = t\,x_1 + \big(1 - (1 - \sigma_{\min})t\big)\,x_0$$

$$\tfrac{d}{dt}\psi_t(x_0) = x_1 - (1 - \sigma_{\min})\,x_0$$

$$x = \psi_t(x_0) \;\Rightarrow\; x_0 = \frac{x - t\,x_1}{1 - (1 - \sigma_{\min})t}$$

$$u_t(x \mid x_1) = \frac{x_1 - (1 - \sigma_{\min})\,x}{1 - (1 - \sigma_{\min})\,t}$$

Global field and trajectory velocity are two representations of the same object.

Optimal Transport Conditional Paths

Choose linear schedules:

MEAN SCHEDULE

$$\mu_t(x_1) = t\,x_1$$

STD-DEV SCHEDULE

$$\sigma_t(x_1) = 1 - (1 - \sigma_{\min})\,t$$

$\Rightarrow$ **leads to the simple vector field derived previously**

The OT Flow Map

THE FLOW MAP

$$\psi_t(x) = \big(1 - (1 - \sigma_{\min})t\big)\, x + t\, x_1$$

STRAIGHT-LINE INTERPOLATION

Each point follows a constant-velocity line from noise toward the sample.

CONTROLLED SHRINKAGE

The noise component contracts smoothly as t increases toward 1.

OT Interpretation

This flow is the Optimal Transport displacement map: mass is moved along geodesics in Euclidean space.

$$p_t = \left[(1-t)\,\mathrm{id} + t\,\psi\right]_{\#}\, p_0$$

KEY PROPERTIES

- ✓ Particles move in straight lines.
- ✓ The transport direction is constant along the path.

Key Insight: Simplest Motion

OT paths define the simplest possible motion between distributions.

GEOMETRY

Samples travel along straight lines, avoiding curved or diffusive trajectories.

VELOCITY

The transport direction stays constant along each conditional path.

LEARNING

Flow Matching learns a more direct field from an easier regression target.

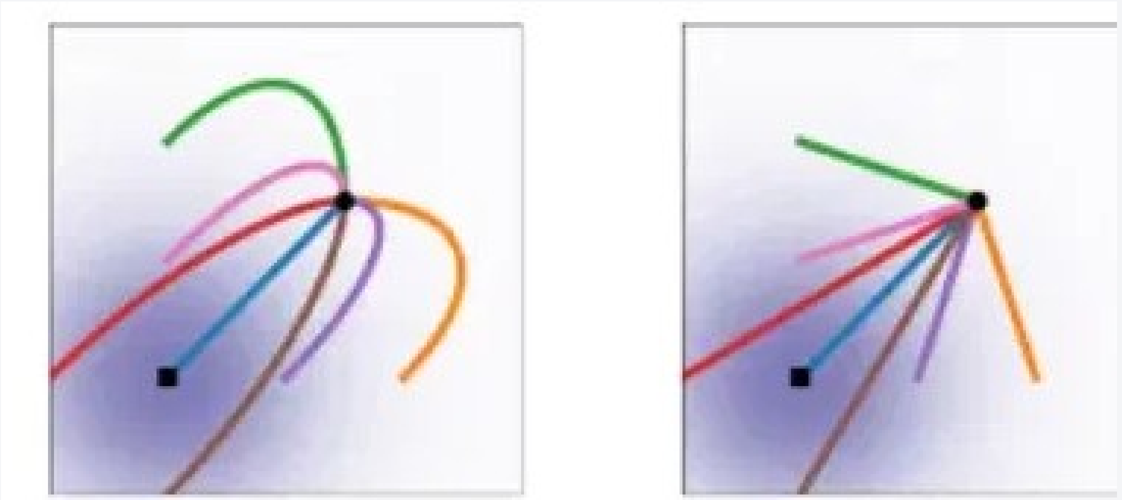
Final Training Objective

CFM LOSS FUNCTION

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}\left[\left\| v_t(\psi_t(x_0)) - \left(x_1 - (1 - \sigma_{\min})x_0\right)\right\|^2\right]$$

A single MSE regression onto a known, closed-form target velocity.

Visual Intuition



COMPARING TRAJECTORIES

- → Left (Diffusion): curved trajectories that bend toward the target.
- → Right (Optimal Transport): straight, direct paths to the target.

Why OT Paths Are Better

Compared to traditional diffusion models:

DIFFUSION PATHS

- ✗ Curved, complex trajectories.
- ✗ Can overshoot the target.

OPTIMAL TRANSPORT

- ✓ Straight-line motion.
- ✓ No backtracking.
- ✓ Simpler, stable dynamics.

Diffusion = density-driven flow · OT = geometry-driven transport

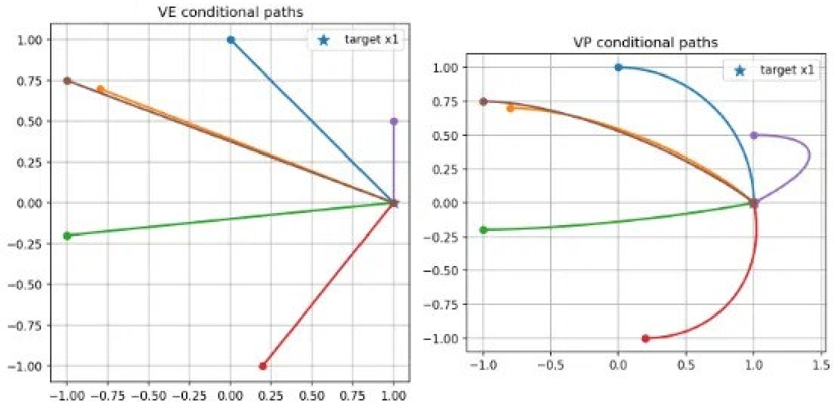
VP Rotation and Conditional Paths

VP FIELD ROTATES OVER TIME

$$u_t(x) \;\propto\; a(t)\,x - b(t)\,x_1$$

- The coefficients a(t) and b(t) change over time.
- Early on one term dominates; later the other does.
- The direction rotates $\Rightarrow$ curved paths.

VE VS VP CONDITIONAL PATHS



VE: straight lines to the target.   VP: rotation and curvature near it.

Final Takeaway

FLOW MATCHING UNIFIES

+ Diffusion models (VE / VP) — recovered as special cases.
+ General probability paths — including Optimal Transport.

KEY IDEA

Learn vector fields directly from chosen probability paths.

PART 07

Results from the Paper

Lipman et al., 2022 — Flow Matching for Generative Modeling

ImageNet & CIFAR-10 · Sampling efficiency · Super-resolution

Experimental Setup

DATASETS

- CIFAR-10.
- ImageNet at 32×32, 64×64, and 128×128.
- Unconditional generation (plus a super-resolution study).

METRICS

- NLL — negative log-likelihood, in bits/dim (↓).
- FID — Fréchet Inception Distance (↓).
- NFE — number of function evaluations (↓).

PROTOCOL

- One shared U-Net (Dhariwal & Nichol) across all losses.
- dopri5 adaptive solver, tolerance 1e-5.
- FID / NFE averaged over 50k samples.

Ablation: same model, swapping the loss — DDPM · Score Matching · ScoreFlow · FM (diffusion) · FM (OT).

Result: Likelihood, Quality & Sampling Cost

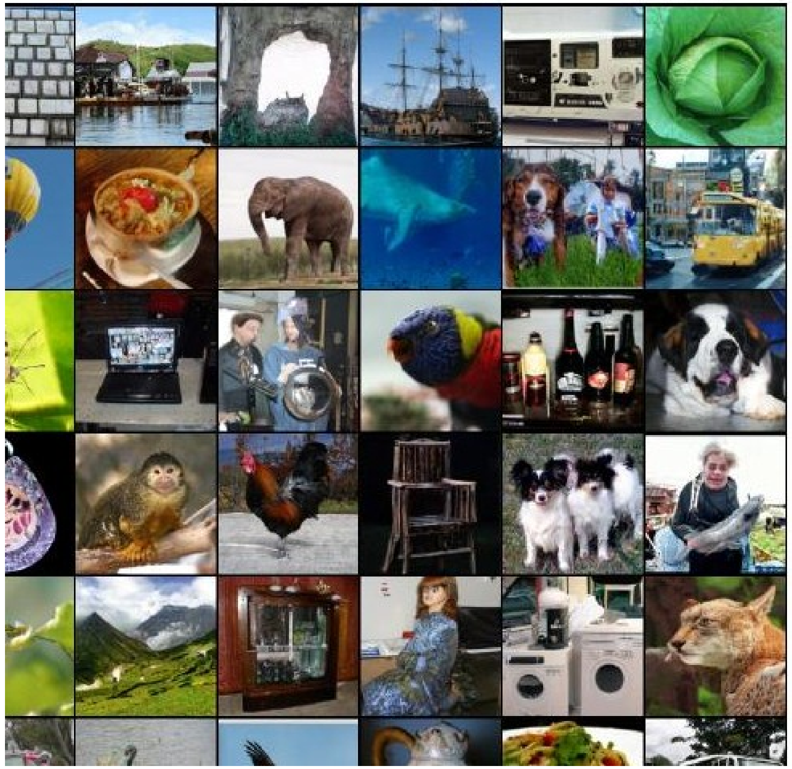
Same architecture, different training losses — lower is better on every metric.

Model	CIFAR-10			ImageNet 32×32			ImageNet 64×64		
	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓
DDPM	3.12	7.48	274	3.54	6.99	262	3.32	17.36	264
Score Matching	3.16	19.94	242	3.56	5.68	178	3.40	19.74	441
ScoreFlow	3.09	20.78	428	3.55	14.14	195	3.36	24.95	601
FM w/ Diffusion	3.10	8.06	183	3.54	6.37	193	3.33	16.88	187
FM w/ OT	**2.99**	**6.35**	**142**	**3.53**	**5.02**	**122**	**3.31**	**14.45**	**138**

FM with OT wins across the board: best NLL, FID and NFE on CIFAR-10 and ImageNet 32/64 — fewer solver steps for better samples and likelihood.

[1] Lipman et al., 2022 · baselines: [8] DDPM, [9] Score Matching, [10] ScoreFlow

Result: ImageNet 128×128 — Near State of the Art



Model	NLL↓	FID↓
Self-cond. GAN	–	41.7
Uncond. BigGAN	–	25.3
PGMGAN	–	21.7
FM w/ OT	2.90	20.9

State-of-the-art FID among comparable methods — and the only one here that also reports a likelihood.

Result: Faster, Cheaper Training

IMAGENET 128×128 TRAINING BUDGET

4.36M → 500K

iterations vs Dhariwal & Nichol — about 33% less image throughput, even with a 25% larger model.

CONVERGENCE

- FM-OT lowers FID faster and further than DDPM, SM and ScoreFlow.
- Baselines such as ScoreFlow / VDM report 1.3M–10M iterations.

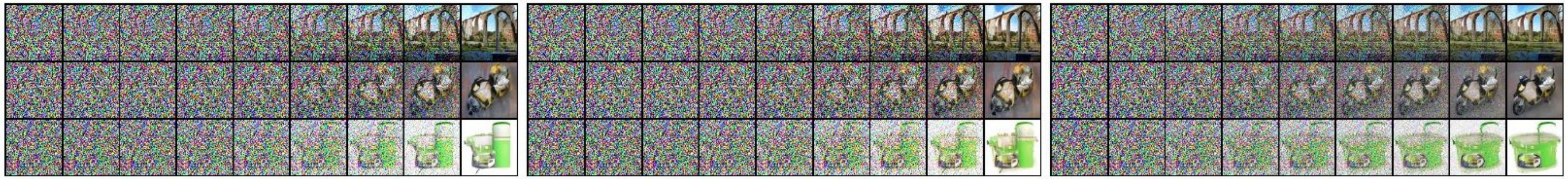
SAMPLING COST STAYS CONSTANT

With Flow Matching the per-sample NFE is roughly fixed throughout training, whereas for score matching it drifts upward as training proceeds.

[1] Lipman et al., 2022 — Flow Matching for Generative Modeling

Result: Straighter Sample Paths

SAME INITIAL NOISE, IMAGENET 64×64 · SCORE MATCHING · FM-DIFFUSION · FM-OT

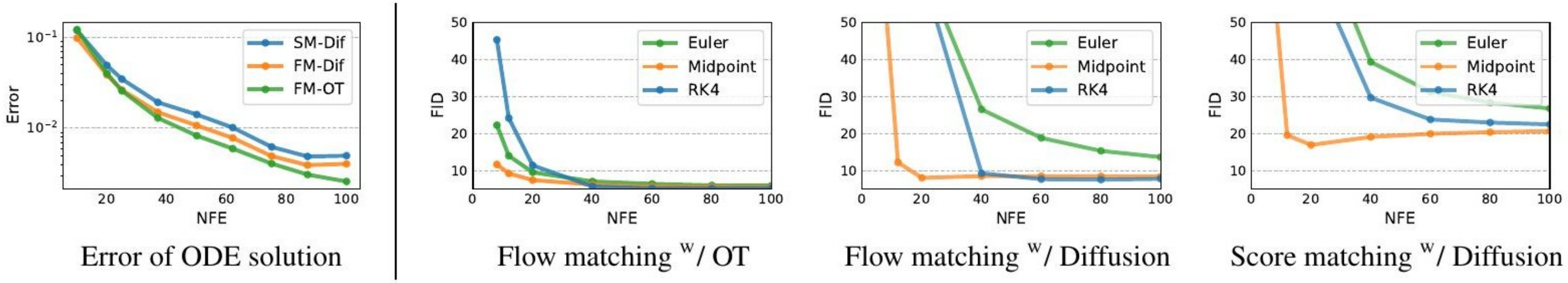


OT reduces noise roughly linearly: the OT-path model starts forming the image early in the trajectory, while diffusion paths keep it buried in noise until the very last steps.

[1] Lipman et al., 2022 — Flow Matching for Generative Modeling

Result: More Efficient Sampling

IMAGENET 32×32 · ERROR AND FID VS NUMBER OF FUNCTION EVALUATIONS



✓ FM-OT reaches the same numerical error using roughly 60% of the NFEs of diffusion models.

✓ It attains good FID even at very low NFE — the best quality-vs-cost trade-off, across every ODE solver.

[1] Lipman et al., 2022 — Flow Matching for Generative Modeling

Result: Conditional Super-Resolution

Upsampling ImageNet images from 64×64 to 256×256 — evaluation protocol of Saharia et al. (2022).

Model	FID↓	IS↑	PSNR↑	SSIM↑
Reference	1.9	240.8	–	–
Regression	15.2	121.1	27.9	0.801
SR3 (Saharia et al.)	5.2	180.1	26.4	0.762
FM w/ OT	3.4	200.8	24.7	0.747

FM-OT improves FID and IS over SR3 with comparable PSNR / SSIM — and FID / IS better reflect perceptual generation quality.

Results — Takeaways

WHAT THE EXPERIMENTS SHOW

- ✓ Best NLL, FID and NFE across CIFAR-10 and ImageNet-32/64 versus diffusion losses on the same model.
- ✓ Near state-of-the-art FID on ImageNet-128 with substantially less training.
- ✓ OT paths give straight, constant-speed trajectories → fewer NFEs and faster, more reliable sampling.
- ✓ The framework extends cleanly to conditional generation (super-resolution).

OT-path Flow Matching: better samples, better likelihood, cheaper sampling — same architecture.

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